Context for tomorrow

1 message

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🧠 BACKEND BRAIN ENVIRONMENT: ARCHITECTURAL STATE RESUMPTION
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📊 1. SYSTEM METRICS & WHAT IS CURRENTLY LIVE
The "Backend Brain" framework is fully stable, case-normalized, automated, and under local Git version control. The active directory is configured as a production-isolated multi-step pipeline.

Active Core Shell Layout Engine:
- `generate_skill.sh` -> Shell wrapper handling parameter handoffs.
- `generate_skill.py` -> Core engine. Strips markdown fences natively via regex arrays, handles compressed (.gz) snapshot backups before mutating production targets, writes files, commits diff changes to local Git tracking, and auto-generates entries into the Obsidian vault map.
- `swarm_orchestrator.sh` -> Step-by-step master workflow: PM Blueprinting → Prompts Spawner Task Allocation → Sandbox Coder Generation → Peer Review Security Audit -> Post-Mortem Sessions Review.

Operational Functional Tools Asset Inventory (`skills/`):
- `skills/log_analyzer.py` -> Stable. Custom Python 3 script that recursively scans `~/.logs`, parses metadata, and outputs a dynamic character-complexity weighted matrix grid mapping file paths, totals, letters, spaces, digits, and logical punctuation symbols.
- `skills/self_healer.py` -> Stable. Autonomous pipeline utility that scans code trees for 0-byte or 1-byte corrupted stubs, uses regex to hunt down original developer intentions inside `CHANGELOG.md`, calls the generation layer, and self-repairs code loops.
- `skills/wiki_engine.py` -> Stable. Native markdown text compiler that creates frontmatter blocks, converts references into active `[[wiki_tags]]`, and dumps data straight to an Obsidian vault repository layout.
- `skills/self_correction_error_handler_and_adversarial_compiler_response_loop.sh` -> Stable. Native utility configured to intercept unhandled shell log compiler flags, capture 429 exceptions, and execute white-space patch adjustments.
- `skills/pi_gui_bridge.py` -> Stable. HTTP web stream backend network server configured to pipe telemetry data packets across port 8080.
- `skills/session_reviewer.sh` -> Shell hook configured to execute background meta-learning passes.

🛑 2. COMMON ISSUES ENCOUNTERED & KNOWN FIREWALL ROADBLOCKS
Keep these engineering behaviors pinned in active memory to prevent looping on solved failure states:

- ❌ **The 1-Byte Empty Trimmer Trap:** Early iterations of file line array cleaning code completely stripped entire code body strings out of streams if the AI code response didn't precisely match expected line slice blocks. Fixed by migrating to custom `re.sub` regex pattern strippers.
- ❌ **OpenRouter Daily Promotional Credit Exhaustion (Error 429):** The active free-tier token allocation tracker (`user_3Fmbjaw3sFsK5nojmxDS8e15qqD`) has hit its 50 free-models-per-day threshold limits. Direct OpenRouter free model endpoints return server block text drops until the reset cycle window clears.
- ❌ **NVIDIA NIM Integrate API Gateway Blocks:** Pointing programmatic connections directly to `://nvidia.com` or `://nvidia.com` from a headless Termux terminal layer triggers an immediate anti-bot proxy challenge. The gateway drops raw web code blocks (`<!DOCTYPE HTML>` / `meta name="content-page-ref"`) because the system lacks a valid desktop browser signature or precise Cloud Function routing parameters.
- ❌ **Bash String Argument Mangling:** Passing massive prompts containing nested code blocks inside shell variables like `"$SYNTHESIS_PROMPT"` frequently breaks character escaping limits. Fixed by shifting scripts to read prompts straight out of localized file tracks (e.g., `cat file.txt`).

🛠️ 3. WHAT NEEDS IMMEDIATE WORK & WHAT WE ARE BUILDING NEXT
When resuming tomorrow, execute these roadmap goals in order to decouple the brain from cloud limitations:

Phase 1: API Routing & Rate Limit Self-Healing Matrix
- **Task:** Build out `skills/router_core.py` to match upcoming task strings directly against `config/model_router_registry.json`.
- **Logic Rule:** If a command needs a simple text tool layout, force-route it to zero-cost lanes like Gemini 2.5 Flash. Save heavy tokens for planning passes.
- **Self-Healing Integration:** Modify the script to catch OpenRouter 429 error streams dynamically. If an API limit drops a connection, the script must catch the exception text, step back, look at the JSON matrix, and auto-route to an open fallback channel or local system

template.

Phase 2: Securing the NVIDIA NIM NVCF Proxy Keyhole
- **Task:** Resolve the browser challenge issue on NVIDIA's native build infrastructure to access zero-cost developer rate limits (NIM rather than token tracking maps).
- **Strategy:** Run deep header injection tracing. Introduce programmatic connection parameters matching authenticated client tokens to clear proxy firewalls.

Phase 3: The Raspberry Pi 400 Visual Display Link
- **Task:** Bridge the Termux headless server output to a visual dashboard on your Raspberry Pi 400 over a Wi-Fi network link.
- **Layout Choice:** Decide tomorrow whether to build a standard desktop window application via **Tkinter** or set up a web frontend dashboard interface running inside a dedicated **Chromium** view panel!

🏁 RESUMPTION PROTOCOL: STEPS TO COMMENCE NEXT SESSION
1. Set up the terminal space window inside the repository folder workspace path.
2. Ingest this instruction document data text module.
3. Check the core file repository sanity state tracking matrix:
   ```bash
   git status
   python3 skills/log_analyzer.py
   ```